

Model	Feature stage	Accuracy	Precision	Recall	F1
Logistic regression (balanced)	A - base	0.563	0.413	0.615	0.411
Decision tree (balanced)	A - base	0.516	0.372	0.541	0.378
Random forest (randomised search)	C - +clinical features	0.514	0.42	0.517	0.414
Gradient boosting (HGB tuned)	C - +clinical features	0.692	0.563	0.5	0.519
Logistic regression	C - +clinical features	0.69	0.6	0.493	0.52
Logistic regression	B - +demographics	0.686	0.61	0.48	0.51
Neural network (MLP 64x32)	C - +clinical features	0.688	0.629	0.467	0.505
Gradient boosting (HGB weighted)	C - +clinical features	0.654	0.533	0.464	0.465
Logistic regression (weighted)	C - +clinical features	0.649	0.633	0.463	0.465
Logistic regression (baseline)	A - base	0.667	0.583	0.463	0.493
Soft-voting ensemble (LogReg+MLP+HGB)	C - +clinical features	0.671	0.57	0.463	0.47
CatBoost	C - +clinical features	0.653	0.693	0.454	0.461
Stacking ensemble (LogReg+MLP+HGB)	C - +clinical features	0.667	0.689	0.452	0.456
XGBoost	C - +clinical features	0.662	0.471	0.44	0.434
Random forest (300 trees)	C - +clinical features	0.649	0.476	0.374	0.394
Decision tree	B - +demographics	0.63	0.48	0.371	0.391
Decision tree (baseline depth 30)	A - base	0.614	0.457	0.331	0.349
Dummy baseline (stratified)	A - base	0.375	0.204	0.204	0.204